

Varun Sharma

Software Engineer — Full Stack & DevOps

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PROFESSIONAL SUMMARY

- Software Engineer with 2+ years of experience in full-stack development, DevOps automation, and AI/ML systems integration. Expertise in designing multi-tenant architectures with hierarchical RBAC, building scalable microservices, and implementing end-to-end CI/CD pipelines. Proficient in cloud infrastructure (AWS, GCP, Azure) using Terraform, container orchestration, and observability tooling. Specialized in RAG-based AI systems, LLM orchestration, and production-grade API optimization with multi-layer caching strategies.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript (ES6+), TypeScript, C++, Kotlin, Bash
- Frontend:** React.js, Next.js, SvelteKit, Tailwind CSS, Material-UI, Puppeteer (Browser Automation)
- Backend:** Node.js, Express.js, FastAPI, Django, RESTful APIs, GraphQL, Microservices, Event-Driven Architecture
- Databases:** PostgreSQL, MongoDB, Redis, Weaviate (Vector DB), Firebase, Query Optimization, Indexing Strategies
- DevOps & Cloud:** AWS (EC2, S3, Lambda, RDS, ECS), GCP (Cloud Run, Compute Engine, Cloud Functions), Azure, Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, Nginx, Celery, RabbitMQ
- AI/ML Stack:** LangChain, LangGraph, RAG Pipelines, Hugging Face, OpenAI APIs, MCP (Model Context Protocol), DSPy
- Observability:** Prometheus, Grafana, Langfuse, ELK Stack, Distributed Tracing, APM
- Architecture:** System Design, Multi-Tenant RBAC, Caching (L1/L2/L3), API Gateway Patterns, Message Queues

EXPERIENCE

- **NerveSparks Pvt. Ltd. / XSparks.ai** New Delhi, India
Software Engineer Jan 2023 – Present
 - **DevOps & Infrastructure Automation:** Architected and maintained end-to-end CI/CD pipelines using Jenkins and GitHub Actions, reducing deployment time by 60%. Implemented infrastructure-as-code with Terraform across AWS, GCP, and Azure for consistent multi-cloud deployments. Configured Nginx reverse proxy with optimized timeouts, 1GB+ file upload support, and load balancing for 10+ microservices.
 - **Multi-Tenant Architecture & RBAC:** Designed and implemented hierarchical RBAC system using Google Identity Platform with 4-tier access control (Platform → Organisation → Department → Users). Built Firebase SDK integration for React frontend and FastAPI backend supporting seamless tenant isolation and permission management.
 - **Backend Optimization & System Design:** Optimized database queries reducing P95 latency by 40% through indexing strategies and query profiling. Implemented multi-layer caching architecture (L1: In-memory, L2: Redis, L3: CDN) achieving 85% cache hit ratio. Designed Celery-based distributed task queues with RabbitMQ for async job processing handling 10K+ daily tasks.
 - **Observability & Monitoring:** Established comprehensive observability stack with Prometheus metrics, Grafana dashboards, and Langfuse for LLM tracing. Implemented distributed tracing and custom alerting reducing MTTR by 50%. Built real-time monitoring for API health, error rates, and resource utilization.
 - **AI/ML Systems & RAG Architecture:** Developed enterprise-grade RAG services with LangChain and Weaviate vector database, serving multiple client applications. Built MCP (Model Context Protocol) servers for tool integration and AI agent orchestration using LangGraph. Integrated voice-enabled RAG with Google Speech-to-Text/TTS for conversational AI interfaces.
 - **Cloud Run & Container Orchestration:** Deployed 15+ FastAPI microservices to Google Cloud Run with auto-scaling configurations. Troubleshooted container startup failures, port binding issues, and cold start optimizations. Managed Docker containerization with multi-stage builds reducing image sizes by 45%.
 - **Meeting Bot Automation Platform:** Engineered headless browser automation system using Puppeteer for joining Google Meet, Microsoft Teams, and Zoom meetings. Implemented real-time audio capture and processing pipelines for meeting transcription and analysis.
 - **Product Development:** Led development of NerveSparks marketing website using SvelteKit with SEO/AEO optimization. Optimized Iris Offline GPT product for constrained hardware environments with responsive UI and efficient resource utilization.

- **Manav Rachna University** Faridabad, Haryana
Web Developer Intern Aug 2024 – Sep 2024
 - **Student Welfare Platform:** Designed and developed the Student Welfare Department website centralizing information access for 5000+ students and faculty with real-time feedback modules for student-teacher interaction.
 - **Biomedical Research Support:** Contributed to HRV (Heart Rate Variability) biomedical research project under Dean supervision, assisting in data analysis, visualization, and research documentation.

PROJECTS

- **Enterprise RAG Platform:** Built centralized RAG service managing reusable pipelines with multi-provider API key management and configurable endpoints. Implemented evaluation methodologies for retrieval quality and response accuracy. *Tech: FastAPI, LangChain, Weaviate, PostgreSQL, Docker, Langfuse.*
- **AI-Powered Document Processing System:** Developed modular document processing pipeline supporting PDF, DOCX, and image extraction with semantic chunking. Integrated DSPy for optimized retrieval-augmented generation with automatic prompt tuning. *Tech: Python, LangChain, Unstructured, ChromaDB, FastAPI.*
- **Imagify – AI Image Generation Platform:** Built text-to-image generation platform using Stable Diffusion with credit-based Stripe monetization. Implemented Redis caching and concurrent request handling for high-throughput image generation. *Tech: React, Node.js, MongoDB, Redis, Hugging Face, Stripe.*
- **Intelligent Job Application Tracker:** Created job tracking system with LLM-based resume analysis and improvement suggestions. Tracked application outcomes to help users iteratively refine profiles based on success patterns. *Tech: MERN Stack, Serverless Functions, OpenAI APIs, PostgreSQL.*

EDUCATION

- **Manav Rachna International University** Faridabad, Haryana
Master of Computer Application; GPA: 8.1 July 2023 – June 2025
- **Maharshi Dayanand University** Rohtak, Haryana
Bachelor of Computer Application; GPA: 7.2 July 2020 – May 2023